The Impact of Derivative Trading on Firm Market Value: a Panel Data and causality Study

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Abstract

Keywords:

1. Introduction

Derivative instruments are widely used by corporations for hedging and risk management purposes. While theoretical frameworks such as Modigliani-Miller (Modigliani and Miller, 1958) suggest that hedging can enhance firm value by reducing financial distress costs, empirical evidence remains mixed. These challenges limit the reliability of existing findings and their applicability to investment and decision-making.

This study aims to investigate the relationship between derivative trading and firm market value. This research will employ a two-stages methodological approach: a panel data regression using recent US data (2010-2023) to provide evidence on the effect between derivative trading and firm market value. Secondly, this study uses causal forests, a machine learning-based method for causal inference. It estimates the overall average treatment effect (ATE) of derivative trading on firm value and Tobin's Q, and decompose it into heterogeneous treatment effects (HTE) by firm financial ratios. The ATE provides the mean impact, while CATE reveals how the effect differs across subpopulations defined by size, leverage, liquidity, growth opportunities, and international exposure.

Corporate risk management has long been analyzed for its impact on firm market value. From the earliest M&M hypothesis Modigliani and Miller (1958) with assumptions of perfect market. While theoretical and empirical research papers suggest that effective risk management can enhance firm value by reducing financial distress costs, stabilizing cash flows, and improving investment efficiency, existing empirical findings are inconsistent Allayannis and Weston (2001). To be finished... adding results and preliminary methodology

1.1. Practical importance of this research

2. Literature review

This literature review analyzes key findings and methodological approaches in the study of derivatives usage and firm value, with a specific focus on how these studies inform our research on the treatment effects of hedging on corporate valuation. And highlighted the theoretical motivations for corporate hedging, empirical evidence on the derivatives-value relationship, and the methodological innovations—particularly causal forest estimation—that enable a possible analysis of treatment effect heterogeneity.

2.1. Theoretical motivations of firms using derivatives

The Modigliani-Miller theorem states that in a perfect market without taxes, asymmetric information, or transaction costs, capital structure is irrelevant to firm value. However, in reality, corporate risk management can enhance firm value by reducing financial distress costs, tax shield affects, and mitigating agency costs. But in real world, Mayers and Smith Jr (1982); Smith and Stulz (1985) is showing that hedging reduces the expected volatility of cash flows, which lowers the probability of default and the expected costs of financial distress. The expected bankruptcy costs is directly related to the size and risk of a firm's fixed liabilities. Based on this point, firms with high leverage and low liquidity are expected to have stronger incentives to hedge.

Froot et al. (1993) has shown that hedging not only reduces the cost of external financing but also decreases reliance on costly external financial markets. The issue of expensive external financing is expected to be more significant for high-growth firms, which often face greater information asymmetry. Based on this theoretical reasoning, firms with better investment opportunities are therefore expected to benefit more from corporate hedging.

While the research by FROOT et al. (1993) has showed that firms with promising growth prospects (proxied by high Tobin's Q) are more motivated to hedge. Their objective is to safeguard internal capital from cash flow shocks, thereby ensuring the ability to fund valuable investments and avoid the cost of bypassing positive-NPV projects due to financial constraints.

Managerial incentives also lead to hedge. Tufano (1996) proofed that even for risk-averse managers, their incomes are tied to firm's stock price, hedging can reduce stock price variability and, consequently, the volatility of their personal wealth. The greater the proportion of a manager's wealth invested in the firm, the more likely the manager is to implement risk management using derivatives.

What's more, the big companies are more motivated to hedge, according to Nance et al. (1993). Larger firms often have better access to market data, financial expertise, and advanced technologies, which reduces the marginal cost of identifying and implementing hedging strategies. Smaller firms, in contrast, may lack the resources or expertise to implement effective hedging programs. They may also find the fixed costs of hedging prohibitive relative to their transaction volumes. It supports that larger firms are more likely to hedge, consistent with the existence of informational and transactional scale economies.

2.2. Relationships between derivatives and firm values

This groundbreaking research Allayannis and Weston (2001) is considered the introduction of the direct method, which attempts to empirically verify that the use of derivatives is positively evaluated by the market and is significantly correlated with firm value, leading to a substantial number of subsequent studies. They examined the use of foreign currency derivatives in a sample of non-financial firms and how this practice affects firm value, revealing a positive relationship between firm value and hedging. It used derivatives as the independent variable. After controlling for other factors influencing firm value, its relationship with firm value—the dependent variable in the model—is analyzed.

Kuiper (2015) investigated the effects of derivatives usage on firm value and firm risk for nonfinancial German firms. This researched showed a negative effect between total derivative positions and market value (Tobin's Q as well). It argued that German managers have less incentive for higher firm values and less firm risk, due to their lesser degree of equity portions in comparison with USA managers.

Bachiller et al. (2021) used meta analysis of 51 studies and showed that hedging presents an economic advantage for all firms, especially those from common law and developed countries. But it lacks of the causality of how can firms using them to increase their value.

2.3. Causality forests

Hornuf and Schaefer (2025) is showing that causal forests are a method of identifying drivers of heterogeneous treatment effects by partitioning the data into subgroups where the treatment effect varies.

The application of causal forests in this thesis aims to address the challenge of uncovering heterogeneous treatment effects of derivative trading on firm market value. These methods provide a data-driven approach to identify heterogeneous of firms where the impact of derivatives trading significantly differs, without the need to pre-specify interaction terms. For this study, the CATE framework is employed to answer a critical research question: For which types of firms does derivative trading significantly enhance or diminish market value (e.g., Tobin's Q and Market value). The ability of causal forests is to model high-dimensional interactions without requiring pre-specified interaction terms makes them an ideal tool in this context.

Athey et al. (2018) introduced generalized random forests, formalizing the estimation of Conditional Average Treatment Effects (CATE) using causal forests. This paper also highlighted that causal forests are a method of identifying drivers of heterogeneous treatment effects by partitioning the data into subgroups where the treatment effect varies.

The strength of this approach lies in its ability to provide valid and interpretable inferences on the average treatment effect for specific groups, defined by covariate values. For this thesis, the CATE framework will help answer the critical question: For which types of firms does derivative trading significantly enhance or diminish market value (e.g., Tobin's Q and market value). And comparing to the panel regression as a robustness check.

As highlighted by Hornuf and Schaefer (2025), causal forests are particularly suited for financial studies, where high-dimensional data and complex, nonlinear relationships are prevalent. Their ability to address endogeneity concerns and improve causal inferences makes them an effective tool for corporate finance research.

Gulen et al. (2025) is showing the empirical method in finance researching and using a binary variable as a treatment. Numerous "causal trees" are grown on random subsamples of the data. Each tree splits the data based on covariates that maximize the difference in treatment effects between the resulting subgroups. A key feature is "honesty," where the splitting rule is determined on one subsample and the treatment effect is estimated on another, preventing overfitting.

Causal forests are particularly powerful for analyzing high-dimensional and nonlinear contexts, as is often the case in financial studies involving derivative trading. Recent finance-oriented applications have demonstrated the utility of these methods in understanding heterogeneity in policy or market interventions, such as the effects of monetary policy, asset pricing anomalies, and fintech innovations. Building on these advances, this thesis will utilize causal forests to explore the nuanced relationships between derivatives usage and firm market value, while addressing endogeneity concerns.

By employing causal forests, this study seeks not only to identify average effects but also to highlight firm-specific characteristics—such as size, industry, or governance structures—that influence the magnitude and direction of the impact. This approach aligns with the overarching objective of addressing heterogeneity and establishing robust causal inferences in the context of derivative trading and firm value.

2.3.1. Comparative Findings

To fill the gap in the literature, we explore the relationship between the use of derivatives to hedge and firm value considering firm characteristics and macroeconomic conditions such as financial crises, exchange rate fluctuations, and economic growth. As stated in previous studies identify a positive correlation between derivative trading and firm value, while others fail to establish significant causal effects. These contradictions may separated into the following factors:

Endogeneity Issues, decisions on derivative trading may not be entirely exogenous. Firms' choices to engage in derivatives may be influenced by unobserved factors (e.g., managerial preferences, corporate culture), leading to endogeneity problems Ullah et al. (2023). Traditional empirical methods may inadequately address these issues, resulting in biased estimates.

Heterogeneity Issues, the relationship between firm value and derivative trading may vary across firms due to differences in firm size Allayannis and Weston (2001), industry characteristics, or market conditions. For instance, large multinational corporations may use derivatives predominantly for managing currency risk, whereas small firms may focus more on raw material price risks. And different motivations were analyzed by different industries Bartram et al. (2011).

Methodological Limitations, traditional regression analyses and fixed-effects models have limitations in causal inference, especially in the presence of complex endogeneity and heterogeneity or policy interventions. Recent advances in causal inference methods (casual forests) offer new possibilities to address these challenges.

Thus, the motivation of this study lies in overcoming these limitations by applying the latest causal inference methods and tools to deeply investigate the causal effects of corporate derivative trading on market value. This research also aims to uncover the heterogeneous impacts of risk management across firms with different characteristics.

2.4. Research question

Based on the review above, the research question lead is shown below: Is hedging driving companies' market value? How much is it higher than the non-hedging companies. And in what way is hedging effective?

3. Methodology

To be finished.

3.1. Casual forest

3.2. Potential Outcomes Framework

We adopt the Gulen et al. (2025) potential outcomes framework. For each unit i:

- $Y_i(1)$: Potential outcome under treatment ($T_i = 1$)
- $Y_i(0)$: Potential outcome under control ($T_i = 0$)
- Observed outcome: $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$

3.3. Het and MSE estimation

The MSE measures the accuracy of treatment effect predictions:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} \left(Y_i^{\text{obs}} - \hat{Y}_i \right)^2$$

where $\hat{Y}_i = \hat{\mu}(X_i) + T_i \cdot \hat{\tau}(X_i)$, and $\hat{\mu}(x)$ is the estimated baseline outcome.

Treatment effect heterogeneity is quantified by the variance of CATE estimates:

$$\text{HET} = \frac{1}{n-1}\sum_{i=1}^{n}\left(\hat{\tau}(X_i) - \hat{\tau}_{\text{ATE}}\right)^2$$

3.4. Honest estimation

The causal forest employs honest estimation by splitting the data into two parts:

$$\begin{aligned}\mathcal{I} &= \mathcal{I}_1 \cup \mathcal{I}_2\\ \text{Tree structure} &: \text{Learned on } \mathcal{I}_1\\ \text{Leaf effects} &: \text{Estimated on } \mathcal{I}_2\end{aligned}$$

3.5. Causal Forest Model

The causal forest is an ensemble of B causal trees. For each tree $b = 1, \dots, B$:

$$\begin{aligned}\hat{\tau}_b(x) = &\frac{1}{|\{i : X_i \in L_b(x), T_i = 1\}|}\sum_{\{i:X_i\in L_b(x),T_i=1\}} Y_i\\ &- \frac{1}{|\{i : X_i \in L_b(x), T_i = 0\}|}\sum_{\{i:X_i\in L_b(x),T_i=0\}} Y_i\end{aligned}$$

where $L_b(x)$ denotes the leaf containing covariates x in tree b.

3.6. Conditional Average Treatment Effect (CATE)

The CATE for a unit with covariates $X = x$ is estimated by averaging over all trees:

$$\hat{\tau}(x) = \frac{1}{B}\sum_{b=1}^{B}\hat{\tau}_b(x)$$

3.7. Average Treatment Effect (ATE)

The ATE is estimated by averaging CATEs over the entire population:

$$\hat{\tau}_{\text{ATE}} = \frac{1}{n}\sum_{i=1}^{n}\hat{\tau}(X_i)$$

3.8. Splitting Criterion

The forest uses a heterogeneity-focused splitting criterion to maximize between-leaf treatment effect differences:

$$\Delta(L_1, L_2) = \frac{|L_1||L_2|}{(|L_1| + |L_2|)^2} \left(\hat{\tau}(L_1) - \hat{\tau}(L_2)\right)^2$$

where L_1 and L_2 are potential child leaves.

4. Sample selection and variable construction

4.1. Sample selection

The sample used in the study consists of about 30000 observations. The following criteria is followed when creating the data set: Corporations with missing data. We investigate this using nearly 70,000 firm-quarter observations of Chinese listed firms from 2000 to 2013. The reasons we select a broad base of Chinese firms are two-fold.

4.2. Variable construction

Tobin's Q is a measure to evaluate a company's investment opportunities. It is calculated by dividing the market value of a company by the replacement value of its assets (Blose and Shieh, 1997).

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